Initialization Sets the Direction of Attention Rank Change During Training

NeuriCo

Abstract

Attention matrices are often said to collapse in rank during training, and spectral statistics are attractive as cheap guides for pruning. Most evidence, however, comes from finished models. We test five specific claims: heads split into high- and low-rank populations, entropy follows layer-specific shapes, heads below $0.3 \log n$ entropy can be pruned safely, $I(A; X) \leq H(Y|X) + O(1/\sqrt{n})$, and the last layer reaches its rank minimum three to five times faster than the first. We train 68 transformers across 13 configurations on text, vision, and pretrained BERT, tracking every head at 41 checkpoints, and run 120 further models with controlled $H(Y|X)$. No run meets the bifurcation criterion under any of five rank definitions. Under default initialization attention starts nearly uniform and rank rises. With label signal, rank falls substantially only when attention starts peaked, and then the last layer is 1.8 times faster than the first (95% CI 1.6–2.0), not three to five times. The entropy threshold removes 63–96% of heads on AG NEWS and CIFAR-10 at a cost of 19–45 percentage points. We give a counterexample to the information bound and prove a valid replacement based on attention variance. Rank collapse is therefore a property of the initial state, not a general law of training. Entropy helps rank heads for pruning on vision tasks, but a fixed threshold is unsafe.

1 Introduction

Does training make attention low-rank? Many practitioners assume that it does, and the assumption has practical weight. Spectral statistics are cheap to compute. If they predicted the fate of a head early in training, they could guide head pruning and low-rank acceleration without any loss-based search.

The evidence for this picture comes mostly from finished models. Studies of head importance measure trained networks [Michel et al., 2019, Voita et al., 2019], spectral studies examine pretrained models or the initial state [Wang et al., 2020, Nait Saada et al., 2025], and the trajectory between these two points is rarely observed. On the theory side, statements about information in attention must also face a known problem: for a deterministic network, mutual information with the input is either constant or infinite [Saxe et al., 2018, Goldfeld et al., 2019].

We focus on a concrete hypothesis made of five claims, listed in table 1. We fixed a numeric criterion for each claim before training any model. **What does "rank collapse" mean if attention starts low-rank?** This question turned out to be central. Scaled dot-product attention [Vaswani et al., 2017] with small query and key weights is close to uniform at initialization, and a uniform matrix has rank one. We therefore track both rank and row entropy, which separate uniform attention from pointer-like attention.

We train 68 transformers in 13 configurations (AG NEWS, SST-2, CIFAR-10, and pretrained BERT [Devlin et al., 2018]), varying depth, normalization placement, label signal, and initial attention sharpness. We record every head at 41 checkpoints. A further 120 runs control $H(Y|X)$ exactly through label noise.

Claim	Verdict	Key evidence
H1: bifurcation by 20% of training	Refuted	0 of 68 runs meet the criterion under any of 5 rank definitions
H2: early monotone fall, late U-shape	Refuted (default init)	Early fall in 0% of default-arm layers; late U in 0–5%
H3: prune $S < 0.3 \log n$ for < 2 pp loss	Refuted in general	Passes in 1 of 11 signal configurations, no better than random
H4: $I(A; X) \leq H(Y\|X) + O(1/\sqrt{n})$	Refuted	Section 4.4 gives a counterexample; the implied constant grows with n
H5: last layer 3–5$\times$ faster to minimum	Refuted; direction only	Undefined under default init; $1.8\times$ [1.6, 2.0] under large init

Table 1: The five claims and our verdicts. A claim counts as supported in a configuration if its prespecified criterion holds in at least 80% of seeds.

Four claims fail as stated, and the pruning claim holds only in narrow conditions. No run shows bifurcation. With default initialization, von Neumann entropy rises: on AG NEWS, 77–86% of the total rank increase occurs in the first 10% of training. With label signal, rank falls substantially only from a peaked start, where the final layer loses 75% of its rank and the first layer 7%. The threshold rule prunes 84.9% of heads on the six-layer AG NEWS model at a cost of 19.3 pp. The information bound is false, and the measured information does not approach it as n grows.

Contributions.

- We measure per-head spectral trajectories across text, vision, depth, normalization, and initialization, and we show that the initial state sets the direction of rank change.
- We refute a proposed information bound with a counterexample, and we prove a replacement in which attention variance relative to noise takes the place of $H(Y|X)$.
- We show that a fixed entropy threshold is unsafe for pruning, while entropy remains a useful ranking signal on CIFAR-10.
- We report what survives: a strong depth gradient in the size of the rank fall from peaked starts.

Organization. Section 2 reviews related work, section 3 defines our measures and criteria, section 4 presents results claim by claim, section 5 interprets them, and section 6 concludes.

2 Related Work

Head importance and pruning. Michel et al. [2019] rank heads by the expected sensitivity of the loss to a head mask. They report that up to 60% of heads can be pruned from a fine-tuned BERT on SST-2, and that the safe share depends on task and model. Voita et al. [2019] prune heads with stochastic L_0 gates and remove 38 of 48 encoder heads for a 0.15 BLEU drop on one translation task. Neither work uses a spectral threshold. Unlike these methods, the rule we test needs no gradients or gates, and we compare it with same-count baselines.

Spectra and rank collapse. Wang et al. [2020] report long-tailed spectra of the attention context mapping in pretrained models, more skewed in higher layers. Nait Saada et al. [2025] analyze rank collapse in attention at initialization and link it to a spectral gap. Definitions of effective rank vary [Roy and Vetterli, 2007], so we check several. Building on this work, we follow the spectrum through training and vary the initial state directly.

Information in deterministic networks. For deterministic networks, mutual information with the input is constant for discrete inputs and infinite for continuous inputs [Saxe et al., 2018, Goldfeld et al., 2019]. Any finite estimate therefore reflects a regularizer. We add a Gaussian noise channel and use pairwise-distance bounds [Kolchinsky and Tracey, 2017], which bracket the regularized quantity. We know of no attention-specific bound of the form in H4, and no derivation of it was available to us. We therefore test the statement itself.

Configuration	Data	Layers	Seeds	Steps	Test acc. (%)
ag_L3/L6/L12	AG NEWS, mean n 46	3 / 6 / 12	5 / 8 / 5	5,526	91.8 / 92.1 / 92.1
sst_L6	SST-2, mean n 22	6	8	3,156	81.0
cifar_L3/L6/L12	CIFAR-10, $n = 65$	3 / 6 / 12	5 / 8 / 5	7,480	79.5 / 81.4 / 81.9
ag_L6_postln	AG NEWS, post-LN	6	5	5,526	92.1
ag_L6_bigqk, cifar_L6_bigqk	Q, K init $\times 3$	6	5 / 5	as above	91.3 / 78.0
ag_L6_shuffled, cifar_L6_shuffled	Shuffled train labels	6	3 / 3	as above	24.7 / 11.3
bert_ag	BERT fine-tune, 40k AG NEWS, 1 epoch	12 (12 heads)	3	1,250	92.5

Table 2: Training configurations, 68 runs in total. Accuracies are seed means. All other models are pre-LN unless stated, and all use 41 checkpoints, dense over the first 20% of training.

3 Methodology

3.1 Definitions

A head's attention $\boldsymbol{A}(x)$ is an $n \times n$ row-stochastic matrix, where n counts non-padding tokens including CLS. With singular values σ_i, we define $p_i = \sigma_i^2 / \sum_j \sigma_j^2$, which are the eigenvalues of $\rho = \boldsymbol{A}\boldsymbol{A}^\top / \operatorname{Tr}(\boldsymbol{A}\boldsymbol{A}^\top)$. Our measures are:

- **Von Neumann entropy:** $S = -\sum_i p_i \log p_i$, with range $[0, \log n]$.
- **Effective rank:** $\mathrm{rank}_{\mathrm{eff}} = \exp(S)$ is our primary measure. We also check stable rank, intrinsic rank $(\sum \sigma_i)^2 / \sum \sigma_i^2$, the Roy–Vetterli rank $\exp H(\sigma / \sum \sigma_i)$ [Roy and Vetterli, 2007], and numerical rank ($\sigma_i > 0.01\,\sigma_1$).
- **Row entropy:** the mean Shannon entropy of rows divided by $\log n$. Uniform attention and pointer attention to a single token both have rank one, but their row entropies are 1 and 0.

A head's value is its mean over 256 fixed development inputs in eval mode. Note that uniform attention has $S = 0$, so a near-uniform head is low-rank.

3.2 Models and data

We use a custom encoder with $d = 192$, 8 heads, GELU, and CLS pooling, trained with AdamW, warm-up, cosine decay, and gradient clipping at 1.0. Table 2 lists the configurations. AG NEWS [Zhang et al., 2015] is truncated to 64 tokens. SST-2 [Socher et al., 2013] trains on phrases and tests on sentences; official test labels are hidden, so test data are half of the 872 labeled validation sentences. CIFAR-10 [Krizhevsky, 2009] uses 4×4 patches, as in a small vision transformer [Dosovitskiy et al., 2021], giving $n = 65$. Two control arms multiply query and key weights by 3 at initialization. This arm was added after the first runs to test whether the absence of collapse is an artifact of small weights, so it is exploratory. Shuffled-label runs reach chance, as they should.

3.3 Prespecified criteria

H1. At 20% of training: at least 10% of heads above $0.6n$, at least 10% below $0.1n$, at least 80% in their union, and a Hartigan dip test [Hartigan and Hartigan, 1985] with $p < .05$.

H2. Early and late mean the first and last thirds of layers. A monotone fall needs Spearman $\rho(t, S) \leq -0.8$ and rebound at most 5% of the range. A U-shape needs a minimum in 5–80% of training, drop and rebound each at least $0.02 \log n$, and rebound at least 25% of the drop.

H3. We remove heads with mean $S < 0.3 \log n$ by zeroing their output, with no rescaling or fine-tuning. The test-accuracy loss must be under 2 pp with its upper 95% bound under 2 pp, and at least 5% of heads must be removed. **Baselines** use the same head count: 10 random draws, highest entropy first, and an ablation oracle based on single-head accuracy drop on development data. A secondary arm fine-tunes for 5% of training steps at a tenth of the learning rate.

H4. The regression $\hat{I} = a + bH(Y|X) + c/\sqrt{n}$ must give a confidence interval for b that includes 1 and a small a, and the excess over $H(Y|X)$ must shrink like $1/\sqrt{n}$. The claim does not define n, so we read it as training-set size.

H5. The time $t_{\min}$ is the first checkpoint within 5% of the fall from the start to the minimum of $\mathrm{rank}_{\mathrm{eff}}$. The ratio of first to last layer must overlap $[3, 5]$, with a fall of at least 10% in both layers.

Configuration	Share $> 0.6n$	Share $< 0.1n$	Between	Median $\mathrm{rank}_{\mathrm{eff}}/n$	Mean dip p
ag_L6	0.000	1.000	0.000	0.060	.63
sst_L6	0.000	0.000	1.000	0.144	.83
cifar_L6	0.000	0.997	0.003	0.036	.74
ag_L6_bigqk	0.000	0.012	0.988	0.200	.35
bert_ag	0.030	0.778	0.192	0.047	.99

Table 3: Head populations at 20% of training, averaged over seeds. H1 needs at least 10% of heads in both of the first two columns; no configuration comes close.

3.4 Estimating information

We run 120 models (4 layers, $d = 128$, 3,000 steps, 3 seeds) over label noise $\eta \in \{0, .1, .2, .3, .4\}$, $n \in \{500, 2\text{k}, 8\text{k}, 32\text{k}\}$, and two tasks. In ORDER, a synthetic task, 32 tokens contain two markers and the label says which comes first, so noise-free $H(Y|X) = 0$ exactly. On AG NEWS we assume noise-free $H(Y|X) \approx 0$. Symmetric noise on training labels gives $H(Y|X) = h_2(\eta) + \eta \log(K-1)$. Clean test data supply the inputs.

Because $\boldsymbol{A}$ is deterministic, we estimate $I(\boldsymbol{A} + \epsilon Z; X)$ over $M = 2{,}000$ test inputs with $\epsilon \in \{0.2, 0.5, 1, 2, 4\}$. The pairwise bounds [Kolchinsky and Tracey, 2017] satisfy $\hat{I}_{\mathrm{BD}} \le I \le \hat{I}_{\mathrm{KL}}$, and both are capped at $\log M = 7.6$ nats. Rerunning at $M = 250$ shows that $\epsilon \le 1$ is affected by this ceiling and $\epsilon \ge 2$ is not. We therefore report $\epsilon = 2$. We chose this scale after seeing the data, and table 8 gives all scales. **Unless stated otherwise, estimates treat all 8 heads of a layer jointly** and average over layers. We also report the per-head mean, which is much smaller at a fixed ϵ.

3.5 Statistics and reproducibility

The seed is the unit of inference. Confidence intervals come from 10,000 bootstrap resamples over seeds, so they are coarse at 3–5 seeds. H3 uses one-sided t-tests against 2 pp with Holm correction across configurations. Paired baseline comparisons use Wilcoxon tests at 8 seeds and t-tests at 5 seeds, since a Wilcoxon test cannot go below $p = .0625$ with 5 pairs. A same-seed rerun of ag_L6 reproduced test accuracy and every logged spectral value exactly. Runs used PyTorch 2.15 on shared NVIDIA RTX A6000 GPUs. Shared memory caused 200 out-of-memory restarts, and interrupted jobs restarted from scratch.

4 Results

4.1 H1: heads form one population

No run in any configuration bifurcates. None of the 68 runs meets the criterion under any of the five rank definitions, at 20% of training or at the end (table 3, figure 1). The choice of definition moves the whole population without splitting it: numerical rank puts 99% of SST-2 heads above $0.6n$ and stable rank puts all of them below $0.1n$. The dip test reaches $p < .05$ in 3 of 68 runs (4.4%), about the chance rate, and two of these are shuffled-label runs. Other definitions give 2–4 of 68.

Only BERT has heads above $0.6n$ under $\mathrm{rank}_{\mathrm{eff}}$: 4% at the start and 3% at the end, mostly in layers 1, 3, and 4. These heads come from pretraining, not from this training run. Head ordering settles fairly early. The Spearman correlation between a head's rank at 20% and at the end is 0.70–0.98 in most configurations, with ag_L12 the exception at 0.32. Ordering stabilizes along a continuum, not into two groups.

4.2 H2: entropy rises under default initialization

The hypothesized pairing of an early monotone fall with a late U-shape occurs in no configuration (table 4, figure 3). In default arms, no early layer shows a monotone fall, and at most 5% of late layers show a U-shape. Instead, entropy rises fast. The share of the total rank increase that occurs in the first 10% of training is 77–86% on AG NEWS (all depths and post-LN), 52–60% on SST-2 and on CIFAR-10 at 3 and 6 layers, and 45% on CIFAR-10 at 12 layers.

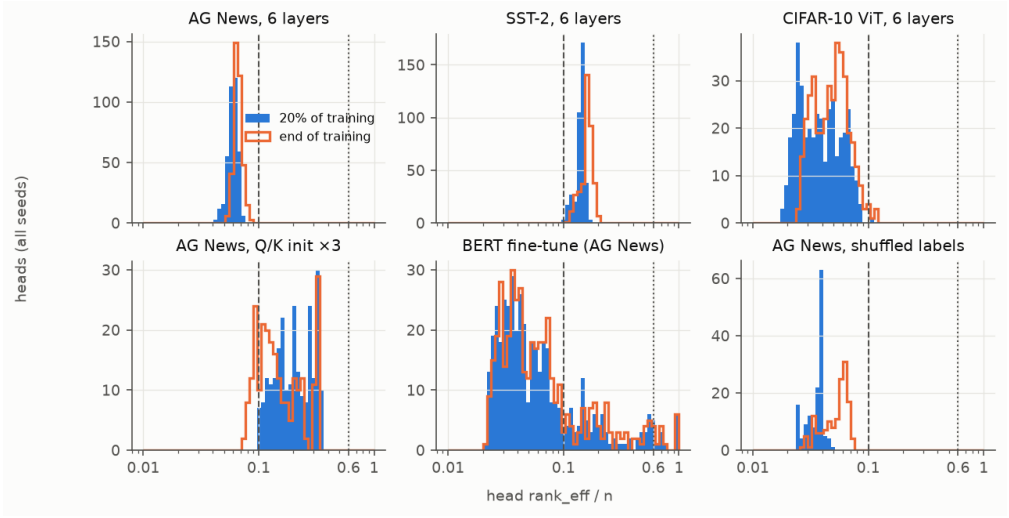


Figure 1: Distribution of head $\mathrm{rank}_{\mathrm{eff}}/n$ at 20% of training and at the end, over all seeds. Dashed and dotted lines mark the claimed limits of 0.1 and 0.6. Heads do not form two groups on either side of these limits; the only heads above 0.6 are in pretrained BERT.

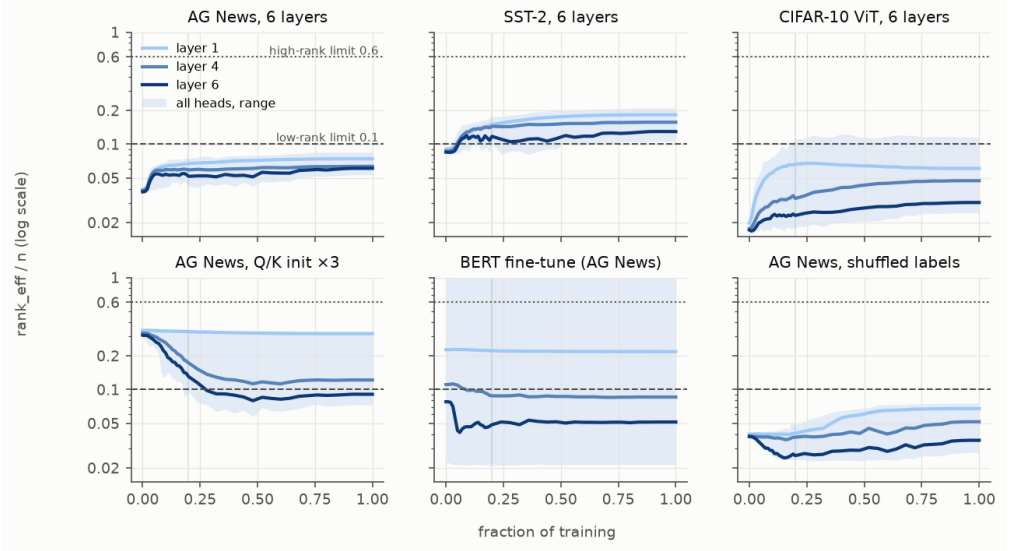


Figure 2: $\mathrm{rank}_{\mathrm{eff}}/n$ over training on a log scale. Thick lines show first, middle, and last layers (seed means), and the band shows the range over all heads and seeds. Rank rises under default initialization (top row). Later layers lose rank from a peaked start (bottom left) and in pretrained BERT, while first layers keep their initial level. Without label signal (bottom right), the last layer dips and then recovers.

A late U-shape does appear in three settings: `cifar_L6_bigqk` (100% of late layers), `ag_L6_shuffled` (100%, 3 seeds), and BERT (67%). It is not tied to large initialization alone, since `ag_L6_bigqk` shows a late fall with no rebound. Rather than a depth law, the U-shape looks like a transient that follows early over-concentration. Early CIFAR-10 layers show the opposite of a U: entropy peaks near 40% of training and then declines slightly.

4.3 H3: the threshold is not a safe pruning rule

The rule passes in only 1 of 11 configurations with label signal (table 5). Across AG NEWS and CIFAR-10 default arms it removes 63–96% of heads at a cost of 19–45 pp. The reason is simple:

	$S/\log n$, start → end		Early layers		Late layers
Configuration	**Early layers**	**Late layers**	**Mono. fall**	**Mono. rise**	**U-shape**
`ag_L6`	.145 → .296	.134 → .263	0%	94%	0%
`sst_L6`	.159 → .394	.143 → .316	0%	100%	0%
`cifar_L6`	.053 → .331	.027 → .172	0%	50%	0%
`ag_L12`	.142 → .274	.118 → .256	0%	35%	5%
`ag_L6_bigqk`	.708 → .636	.686 → .353	50%	0%	0%
`cifar_L6_bigqk`	.367 → .419	.322 → .213	0%	10%	100%
`ag_L6_shuffled`	.146 → .273	.136 → .136	0%	50%	100%
`bert_ag`	.428 → .428	.184 → .204	0%	17%	67%

Table 4: Entropy shapes. Percentages are shares of (seed, layer) cells in the given band of layers. H2 needs a monotone fall early and a U-shape late in the same configuration.

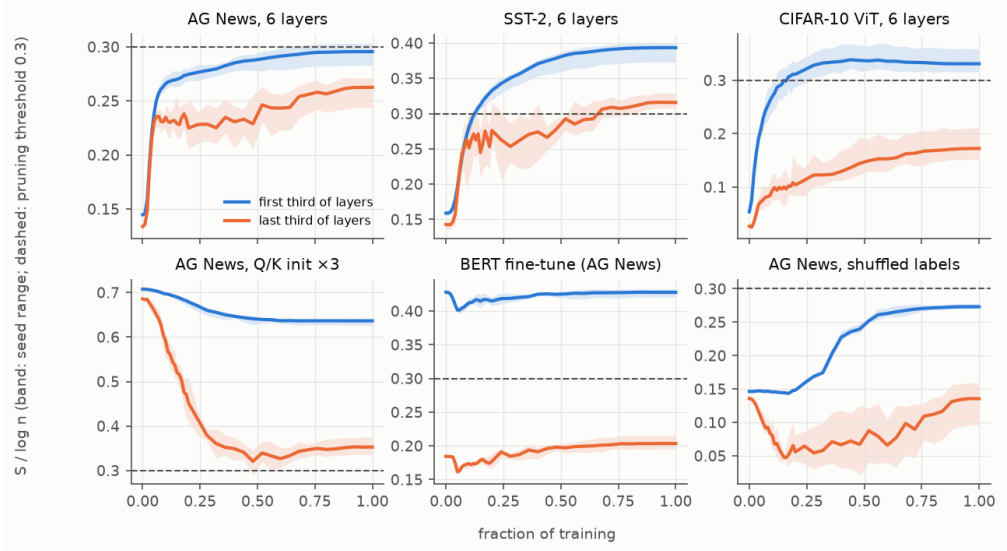


Figure 3: $S/\log n$ for the first and last thirds of layers; bands show the seed range and the dashed line marks the 0.3 pruning threshold. Default runs (top row) rise steeply and end near the threshold, which explains why it splits them so poorly (section 4.3).

attention is near-uniform, so most heads lie below $0.3 \log n$, and the threshold selects most of the model instead of a redundant minority. `ag_L6_bigqk` has a small loss but prunes only 3.8% of heads, which fails our non-vacuity rule.

SST-2. The loss meets the criterion, but entropy adds nothing here: the difference from random is -0.36 pp (Wilcoxon $p = .25$). With 436 test sentences, one sentence is 0.23 pp, so this comparison is weak. SST-2's normalized entropy ends higher than AG NEWS's (.32–.39 against .26–.30), so few heads fall below the line. Its sequences are also about half as long (mean n of 22 against 46), and we did not test whether length is the cause.

Entropy is a real signal on CIFAR-10 but not on text. At the threshold's head count, removing low-entropy heads first beats random on all three default CIFAR-10 depths: by 8.5 pp at $L = 6$ (Wilcoxon $p = .008$), 12.7 pp at $L = 12$ (t-test $p = .005$), and 3.2 pp at $L = 3$ (t-test $p = .002$). Removing high-entropy heads first costs far more, 56.5 pp against 32.8 pp at $L = 6$. The Spearman correlation of S with ablation importance is $+0.23$ to $+0.40$ on CIFAR-10 and -0.14 to $+0.11$ on default-initialization text. On AG NEWS, low-S pruning is if anything worse than random, by 1.4 pp at $L = 6$ (Wilcoxon $p = .016$). On BERT it is worse by 27 pp, though imprecisely (3 seeds, t-test $p = .12$). The ablation oracle does not always win, because single-head importance is a poor guide to joint removal.

Smaller cuts and fine-tuning. When we cut only 25% of heads, default AG NEWS models and SST-2 lose 0.7–1.7 pp under every selection rule. This echoes Michel et al. [2019]: much head

Configuration	Pruned	Low-S loss [95% CI]	Random	High-S	Oracle	Holm p (< 2 pp)
sst_L6	13.8%	**0.12** [−0.43, 0.75]	0.47	1.03	0.55	.004
ag_L6_bigqk (vacuous)	3.8%	**0.12** [0.01, 0.27]	0.24	0.22	0.26	<.001
ag_L3	95.8%	45.3 [42.4, 48.2]	42.1	**39.2**	43.3	1
ag_L6	84.9%	19.3 [16.4, 22.0]	**17.9**	18.4	19.0	1
ag_L12	90.2%	24.1 [15.9, 31.6]	25.8	**23.7**	24.2	1
ag_L6_postln	89.2%	28.6 [20.4, 39.7]	26.2	30.0	**25.1**	1
cifar_L3	63.3%	36.7 [28.2, 44.8]	40.0	47.0	**36.6**	1
cifar_L6	66.9%	**32.8** [28.8, 37.1]	41.3	56.5	35.6	1
cifar_L12	70.0%	**27.6** [23.8, 32.2]	40.3	58.5	32.5	1
cifar_L6_bigqk	46.7%	19.5 [17.5, 21.4]	22.6	46.4	**16.2**	1
bert_ag	64.4%	40.9 [15.3, 58.6]	13.6	25.5	**7.6**	1

Table 5: Accuracy loss in percentage points at $\tau = 0.3$, with all baselines at the same head count. The lowest loss in each row is in **bold**. Only sst_L6 passes with a non-vacuous cut. Shuffled-label runs are omitted.

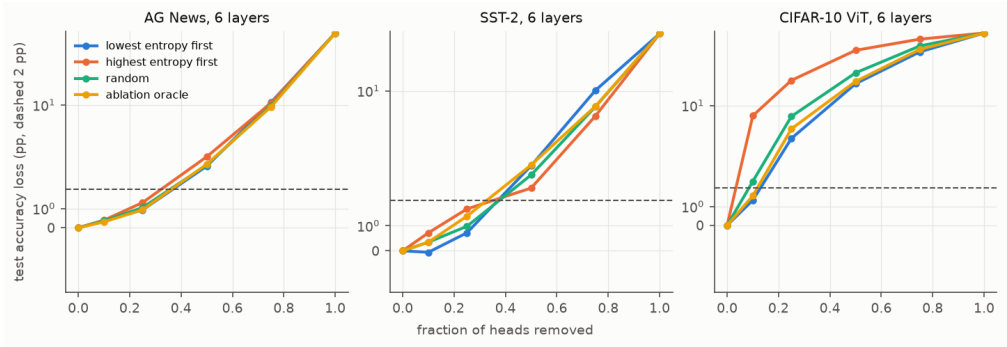


Figure 4: Accuracy loss against share of heads removed for six-layer models, on a symmetric-log axis; the dashed line marks 2 pp. On CIFAR-10, removing high-entropy heads first is far worse than the other rules, while on AG NEWS all rules coincide.

removal is cheap under any method. CIFAR-10 at $L = 6$ is different, losing 4.6 pp with low-S first and 19.8 pp with high-S first. Fine-tuning recovers much of the loss but not enough: AG NEWS goes from 19.3 to 10.4 pp and CIFAR-10 from 32.8 to 8.5 pp, while SST-2 stays at 0.12 pp.

4.4 H4: the bound is false

Proposition 1 (Counterexample)**.** *Let X be uniform on $M \geq 2$ inputs and $Y = f(X)$, so $H(Y|X) = 0$. Take any network that computes f and has a head whose attention map $x \mapsto \boldsymbol{A}(x)$ is injective on these inputs, with that head's output weight set to zero. Predictions are unaffected, and since $\boldsymbol{A}$ is a function of X, $I(\boldsymbol{A}; X) = H(\boldsymbol{A}) = \log M$ for every n. The right side of the claim tends to 0.*

The training-set size cannot rescue the claim: $I(\boldsymbol{A}; X)$ is a property of the map, and nothing forces a Bayes-optimal network's attention to discard input information. The direction is also suspect. The data-processing inequality gives $I(\boldsymbol{A}; Y) \leq I(X; Y) = H(Y) - H(Y|X)$, which falls with $H(Y|X)$, whereas the claim rises with it.

Proposition 2 (Variance bound)**.** *For $T = \boldsymbol{A}(X) + \epsilon Z$ with Z standard Gaussian and $\bar{\boldsymbol{A}} = \mathbb{E}\,\boldsymbol{A}(X)$,*

$$I(T; X) \leq \min\left\{ \log M,\ \frac{\mathbb{E}\|\boldsymbol{A}(X) - \bar{\boldsymbol{A}}\|_F^2}{2\epsilon^2} \right\}.$$

Proof. $I(T; X) = \mathbb{E}\,\mathrm{KL}(\mathcal{N}(\boldsymbol{A}(X), \epsilon^2 I) \,\|\, P_T) \leq \mathbb{E}\,\mathrm{KL}(\mathcal{N}(\boldsymbol{A}(X), \epsilon^2 I) \,\|\, \mathcal{N}(\bar{\boldsymbol{A}}, \epsilon^2 I))$, because the true marginal P_T minimizes the expected KL over all reference distributions. The right side equals the variance term. The $\log M$ term follows from $I(T; X) \leq H(X)$. □

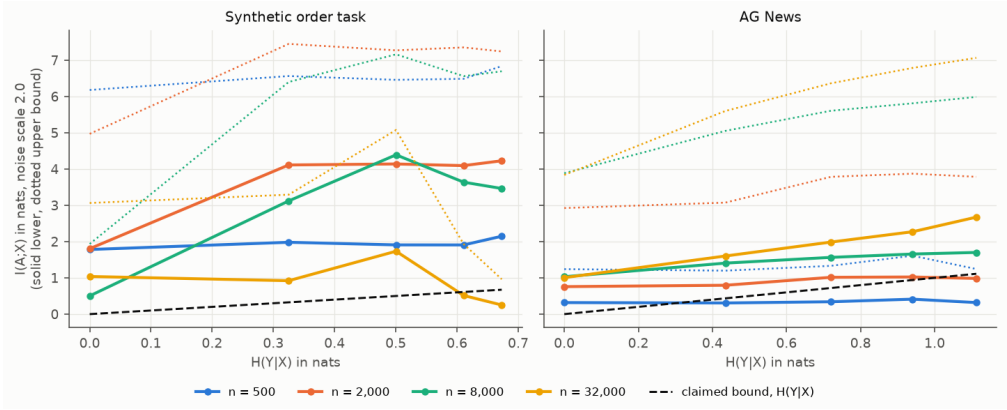


Figure 5: Estimated layer-level information at $\epsilon = 2$, averaged over layers and seeds. Solid lines are the lower bound and dotted lines the upper bound. The dashed black line is the claimed $H(Y|X)$ with the $O(1/\sqrt{n})$ term set to zero. Noise-free points (left end) lie above the line at every n, and the curves do not approach it as n grows.

What bounds the information is attention variability relative to the noise, and $H(Y|X)$ does not appear. $\hat{I}_{\mathrm{BD}}$ stayed below this bound in all 19,200 checks (every layer, noise scale, and checkpoint). $\hat{I}_{\mathrm{KL}}$ may exceed it, since Jensen's inequality only gives $\hat{I}_{\mathrm{KL}} \leq 2\times$ the bound, and it does so at $\epsilon = 2$.

Measurements. Table 6 and figure 5 show noise-free results. At $H(Y|X) = 0$ and 100% accuracy on ORDER ($n = 32{,}000$), layer-level information is at least 1.04 nats on average, where the claim allows only $C/179$. Seeds vary widely here, from 0.21 to 1.83 nats. The per-head mean is 0.13 nats.

The excess does not shrink with n. A true $O(1/\sqrt{n})$ term needs a bounded C. On AG NEWS the implied C rises from 7 to 180 at the layer level and from 0.9 to 23 at the head level. On ORDER it rises non-monotonically, from 40 to 186 and from 5 to 24. The ORDER trend is noisy and its $n = 500$ cell did not learn the task, so AG NEWS carries more of the weight here. The regression also fails at every scale (table 8). Where the interval for b includes 1 in the layer-level fits, the intercept is 1.4–5.5 nats, not near zero. On AG NEWS, c is negative at every scale: information rises with n, the opposite of the claimed term.

Single numbers depend on the level and the noise scale. At $\epsilon = 2$, layer-level $\hat{I}_{\mathrm{BD}}$ exceeds $H(Y|X) + 1/\sqrt{n}$ in 93% of ORDER runs and 73% of AG NEWS runs. For the per-head mean these shares fall to 35% and 15%, and at $\epsilon = 1$ they are 90% and 47%. Large noise can make the claim pass trivially: at $\epsilon = 4$, noise-free ORDER runs at $n = 8{,}000$ reach 0.13 nats. Cells below the claimed value are mostly noisy ones, such as AG NEWS at $\eta = 0.4$, $n = 500$ (0.32 against 1.11 nats), and these are consistent with the claim only because information is low there, not because it tracks $H(Y|X)$. An untrained network already exceeds the claim at low noise (AG NEWS at $\epsilon = 1$: $\hat{I}_{\mathrm{BD}} = 0.65$ nats at step 0). For these reasons we rest the refutation on Proposition 1 and on the trend in n, and not on any one number.

Label noise does raise information, which fits the claim in sign only: at $n = 32{,}000$ on AG NEWS, $\hat{I}_{\mathrm{BD}}$ goes from 1.0 to 2.7 nats as η goes from 0 to 0.4. Memorization of noisy labels is a likely mechanism, though we did not test it. Saturation is partial. At $n = 8{,}000$, AG NEWS plateaus by 40% of training and noise-free ORDER by 20%, while ORDER at $\eta = 0.2$ keeps rising (0.5 nats at 20%, 3.8 at 60%, 4.4 at the end). Plateaus do not sit at $H(Y|X)$.

4.5 H5: depth shapes the size of the fall more than its speed

Under default initialization there is no collapse to time (table 7). The minimum sits at or near the start, because rank rises: $\mathrm{rank}_{\mathrm{eff}}/n$ goes from .04 to .07 on AG NEWS (layer 1), .09 to .18 on SST-2, and .02 to .06 on CIFAR-10. Three signal configurations show a real fall in deep layers. `ag_L6_bigqk` gives the only interpretable ratio, 1.81, which is below the claimed range, and even there layer 1 falls only 7%. In `cifar_L6_bigqk` and BERT, layer 1 hardly falls, so the ratio has

Noise-free runs, $\epsilon = 2$	$n = 500$	$n = 2{,}000$	$n = 8{,}000$	$n = 32{,}000$
ORDER: layer $\hat{I}_{\mathrm{BD}}$, nats (test accuracy)	1.78 (49%)	1.80 (99%)	0.50 (100%)	1.04 (100%)
ORDER: implied C, layer / head	40 / 5	81 / 10	45 / 6	186 / 24
AG NEWS: layer $\hat{I}_{\mathrm{BD}}$, nats (test accuracy)	0.32 (63%)	0.76 (76%)	1.03 (83%)	1.01 (87%)
AG NEWS: implied C, layer / head	7 / 0.9	34 / 4	92 / 12	180 / 23

Table 6: Noise-free information. Implied C is the excess of $\hat{I}_{\mathrm{BD}}$ over $H(Y|X)$ times $\sqrt{n}$, which a true $O(1/\sqrt{n})$ term would keep bounded. "Layer" treats all 8 heads jointly; "head" is the mean over heads.

Configuration	$t_{\min}$ **layer 1**	$t_{\min}$ **last**	**Ratio [95% CI]**	**Rank drop, first → last**	**Reading**
ag_L3, ag_L6, cifar_L3	0	0	undefined	0 (rank rises)	Minimum at initialization
ag_L12, cifar_L6, cifar_L12, sst_L6	≤.001	.010–.124	≈0	≈0	Same
ag_L6_bigqk	.69	.38	**1.81** [1.63, 2.00]	7% → 75%	Real collapse, below 3×
cifar_L6_bigqk	.002	.14	0.01 [0, .04]	0.4% → 60%	Layer 1 never falls
bert_ag	.63	.05	11.9 [10.0, 14.4]	4% → 47%	Layer 1 barely falls; unstable

Table 7: Time to minimum rank as a fraction of training. The criterion needs a fall of at least 10% in both layers. No run in any configuration satisfies this.

little meaning. Without label signal, the last layer of ag_L6_shuffled drops 37% and then recovers, so a transient fall does not need a peaked start.

Post-hoc rise times to peak rank have no consistent sign (figure 6). The mean per-seed ratios of layer 1 to last layer are 7.8, 2.1, and 0.63 on AG NEWS for $L = 3, 6, 12$, and 0.42, 0.26, and 0.21 on CIFAR-10. The value of 2.1 hides heavy tails: its median is 0.78. SST-2 gives 3.7 [2.4, 5.0], which overlaps the claimed range, but this measures a different quantity on a post-hoc basis.

What holds up is the size of the fall. From a peaked start, the rank drop grows steadily with depth: 7, 36, 56, 66, 70, and 75% from layer 1 to 6 on AG NEWS, and 0.4, 14, 30, 42, 53, and 60% on CIFAR-10. BERT's 4% to 47% matches this pattern. In addition, the first layer has higher rank than the last at 95–100% of checkpoints in every configuration, though not always in strict order across intermediate layers. This agrees with Wang et al. [2020], who report more skewed spectra in higher layers.

5 Discussion

Why did the framing fail? The hypothesis imagines full-rank matrices that collapse. With standard initialization, scaled dot-product attention begins near-uniform (row entropy about 0.99) because queries and keys are small, so $\boldsymbol{A}$ is close to a rank-one matrix and learning builds structure. The $\times 3$ arm is the key control. When attention starts peaked (row entropy 0.48, $\mathrm{rank}_{\mathrm{eff}}/n$ about 0.32 on AG NEWS), rank falls, and falls more with depth. Last layers then end close together from either side, at .06 for default and .09 for $\times 3$ on AG NEWS, though first layers keep their starting level (.07 and .32). The start state therefore persists early in the network and matters less late in it. BERT starts peaked as well (row entropy about 0.50), and its last layers lose rank during fine-tuning, which agrees with this reading.

Practical implications. We advise against a fixed $0.3 \log n$ cut-off for pruning. Descriptions of attention training as rank collapse should state the initial state, since it sets the direction. On vision models, entropy is a useful ranking signal, but this does not carry over to text. We did not expect this split. One possible reason is that a fixed image grid gives heads stable spatial roles while text attention varies by input, but we did not test this.

Alternative explanations. *Small models:* BERT is far larger and pretrained, and it is also unimodal and mostly low-rank (mean dip $p = .99$). *Pre-LN:* post-LN matched pre-LN almost exactly. CLS *pooling:* this was constant across arms, so we cannot rule out an effect. *Averaging across inputs:* per-head means could hide bimodality at the level of single inputs, and this is a real limitation.

Limitations. We study small encoders and short fine-tunes on three classification datasets. We did not study ImageNet-scale vision transformers, decoders, or large language models, where bifurca-

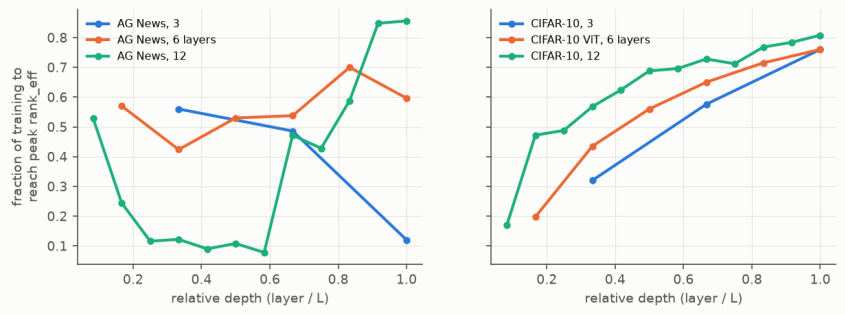


Figure 6: Fraction of training needed for $\mathrm{rank}_{\mathrm{eff}}$ to reach its peak, by relative depth and network size, under default initialization. CIFAR-10 peaks later in deeper layers, while AG NEWS shows no consistent order. The direction of the depth effect depends on the task.

tion might hold, although nothing here suggests it. The definitions of $\mathrm{rank}_{\mathrm{eff}}$ and S were our choice, as the hypothesis gave none, and five rank definitions cover only H1. The $\times 3$ arm, rise-time ratios, $\epsilon = 2$ as reporting scale, and paired baseline tests were not pre-registered, and we tested only one gain value, so these analyses are exploratory.

For H3, SST-2 evaluation has 436 sentences, BERT has 3 seeds with very wide intervals, and we ran fine-tuning only on $L = 6$ default arms. Zeroing a head without rescaling is one of several removal methods.

For H4, noise channels define a quantity only relative to ϵ. Layer-level and per-head values differ by about a factor of eight at a fixed $\epsilon \geq 1$, and the claim does not say which it means. AG NEWS's intrinsic $H(Y|X)$ is assumed near zero and is really some small unknown value. Injected noise applies to training labels, while clean test inputs supply X. On ORDER, training failed at small n (test accuracy near 50% at $n = 500$ and for noisy runs at $n = 2{,}000$), and 3-seed means are noisy. Our reading of n as training-set size is an assumption. A derivation with stronger assumptions, such as stochastic attention or a different left side such as $I(\boldsymbol{A}; Y)$, could survive, but it would not be this claim.

Time to minimum uses a 5% tolerance on 41 checkpoints, so differences below about 1% of training early on, or 4% later, cannot be resolved. Shared-GPU restarts should not affect results, and the exact rerun supports this, but it covered one configuration on one GPU type.

6 Conclusion

We tested five claims about attention spectra over training, with per-head tracking across 68 runs and 120 information runs. Heads do not split into high- and low-rank populations. Under standard initialization they start near-uniform and gain rank, so rank collapse is not a universal training dynamic. The bound $I(A; X) \leq H(Y|X) + O(1/\sqrt{n})$ is false, and our variance bound is a valid replacement. The $0.3 \log n$ pruning rule is unsafe, though entropy carries pruning information on CIFAR-10. One finding survives in modified form: the rank fall from peaked starts grows strongly with depth.

The takeaway: the initial state, not training alone, decides whether attention rank falls.

Future work should sweep query and key gain and softmax temperature to map how the start state carries through depth, test a per-layer entropy ranking on larger vision models, ask whether stable spatial roles explain the link between entropy and importance on vision, and analyze per-input distributions, since per-head means may hide input-level bimodality. If a derivation of the H4 bound exists, its assumptions deserve a direct test.

References

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. arXiv:1810.04805, 2018.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Georg Heigold, Thomas Unterthiner, Marc Brockschmidt, and Frank Hutter. An image is worth 16x16 words: Transformers for image recognition at scale. In *Proceedings of ICLR*, 2021.

Ziv Goldfeld, Ewout van den Berg, Kristjan Greenewald, Igor Melnyk, Nam Nguyen, Brian Kingsbury, and Yury Polyanskiy. Estimating information flow in deep neural networks. In *Proceedings of ICML*, pages 2299–2308, 2019.

J. A. Hartigan and P. M. Hartigan. The dip test of unimodality. *The Annals of Statistics*, 13(1): 70–84, 1985.

Artemy Kolchinsky and Brendan D. Tracey. Estimating mixture entropy with pairwise distances. *Entropy*, 19(7):361, 2017. Corrected version at arXiv:1706.02419.

Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

Paul Michel, Omer Levy, and Graham Neubig. Are sixteen heads really better than one? In *Advances in Neural Information Processing Systems 32*, 2019.

Thiziri Nait Saada, Alireza Naderi, and Jared Tanner. Mind the gap: a spectral analysis of rank collapse and signal propagation in attention layers. In *Proceedings of ICML*, 2025. arXiv:2410.07799.

Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *Proceedings of EUSIPCO*, 2007.

Andrew M. Saxe, Yamini Bansal, Joel Dapello, Madhu Advani, Artemy Kolchinsky, Brendan D. Tracey, and David D. Cox. On the information bottleneck theory of deep learning. In *Proceedings of ICLR*, 2018.

Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of EMNLP*, pages 1631–1642, 2013.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30*, 2017.

Elena Voita, David Talbot, Fedor Moiseev, Rico Sennrich, and Ivan Titov. Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned. In *Proceedings of ACL*, 2019.

Sinong Wang, Belinda Z. Li, Madian Khabsa, Han Fang, and Hao Ma. Linformer: Self-attention with linear complexity. arXiv:2006.04768, 2020.

Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems 28*, 2015.

A Information Regressions at All Noise Scales

B Deviations from the Plan

The plan was frozen before training. We changed the following afterwards: (1) the noise grid moved from $\{.05\text{–}1\}$ to $\{.2\text{–}4\}$ after a pilot showed that small scales pin at the ceiling; (2) we added the $\times 3$ arm after the first 5 runs; (3) we gathered the literature by hand; (4) we added the correlation of S with ablation importance as a descriptive statistic; (5) fine-tuning moved to a separate script that uses saved weights, so it was not available for BERT; (6) the job queue was reordered and GPU reservations lowered for the information and BERT runs; (7) we chose $\epsilon = 2$ after seeing the data; (8) we added rise-time ratios because time to minimum is degenerate under default initialization. All planned seeds finished.

Task	ϵ	a **[95% CI]**	b **[95% CI]**	c	**Mean** $\hat{I}_{\mathrm{BD}}$
ORDER	0.5	5.33 [4.45, 6.20]	0.79 [−0.74, 2.32]	52.2	6.76
	1	3.06 [1.94, 4.19]	2.23 [0.25, 4.21]	71.1	5.50
	2	1.38 [0.71, 2.05]	1.94 [0.70, 3.17]	9.0	2.38
	4	0.36 [0.18, 0.53]	0.50 [0.18, 0.83]	2.1	0.61
AG NEWS	0.5	8.33 [8.01, 8.65]	0.18 [−0.19, 0.56]	−75.4	6.87
	1	5.54 [5.11, 5.97]	1.47 [1.00, 1.95]	−118.4	4.00
	2	1.58 [1.43, 1.74]	0.59 [0.35, 0.82]	−38.3	1.16
	4	0.40 [0.36, 0.44]	0.15 [0.09, 0.22]	−9.9	0.30

Table 8: Layer-level fits of $\hat{I}_{\mathrm{BD}} = a + bH(Y|X) + c/\sqrt{n}$. The $\epsilon = 0.2$ fits are omitted since they sit at the $\log M$ ceiling, and $\epsilon \leq 1$ is ceiling-affected. No scale gives both b near 1 and a near 0. Per-head fits at $\epsilon = 2$ give $a = 0.18$, $b = 0.25$ [0.09, 0.41] on ORDER and $a = 0.20$, $b = 0.08$ [0.05, 0.11] on AG NEWS, so they also fail.